(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

IV Semester, April, 2024

Paper Code: CIC 210 **Subject: Database Management System**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1	a. Differentiate between a strong and weak entity type with an example of each.	2.5	CO1
	b. Differentiate between physical and logical data independence.	2.5	CO1
	c. Explain DDL, DML and DCL in SQL. List the commands under each category.	2.5	CO2
	d. List different types of attributes of an entity with an appropriate example.	2.5	CO1
2	a. Construct an ER diagram for a car-insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents.	5	CO1
	b. Explain the following concepts w.r.t Extended ER diagrams (with an appropriate Example): i) Specialization ii) Aggregation	5	CO1
3	Consider the following relational schema: *employee (employee name, street, city, manager name)* *works (employee name, company name, salary)* *company (company name, city)* write SQL or Relational algebra expression for each of the following queries. i) Find the names of all employees who live in the same city as the company for which they work. ii) Find the names and cities of residence of all employees who work for Small Bank Corporation. iii) Find the total number of employees working in each company. iv) List the name of the employees who are working in the same company where the employee Smith is working. (Note: Smith is the name of an employee) v) Name the employee who is earning maximum salary.	2*5=10	CO2
4.	a. Differentiate between Super Key, Candidate Key and Primary Key with an example.	5	CO2
	b. Explain natural join, left outer join, right outer join w.r.t SQL with example.	5	CO2

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END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] JUNE 2024

Paper Code: CIC-210 Subject: **Database Management System**

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt questions in all including Q.No.1 which is compulsory. Select one question from each Unit.

Q1 a) Give an example of mapping of generalization or specialization into relation schemas Summarize the steps involved in converting the ER constructs to relational schema. **(3)**

b) What is a Weak Entity type? Explain the role of partial key in design of weak entity type. **(3)**

c) Consider a relational schema with the following tables:
Employee (EmpID, EmpName, DeptID)
Department (DeptID, DeptName)
Write a relational calculus expression to retrieve the names of all employees who work in the "Sales" department. **(3)**

d) Why is the domain key normal form (DKNF) known as ultimate normal form? **(3)**

e) Compare and contrast heap files and sorted files in terms of storage efficiency and query performance. When would you choose to use a heap file over a sorted file, and vice versa? **(3)**

UNIT-I

Q2 a) Design an ER — Diagram for a UNIVERSITY database schema taking into account at least five entities and indicate all keys and cardinality constraints and assumptions that are made. **(7)**

b) Define the following with examples: **(5)**
(i) Super key (ii) Candidate key
(iii) Primary key (iv) Foreign key

c) What do you mean by constraints? Differentiate between table and column constraint. Explain with an example. **(3)**

Q3 a) Consider the following tables: **(8)**
works (Pname, Cname, Salary)
lives (Pname, Street, City)
located-In (Cname, City)
write the following queries in SQL:
i) List the names of the people who work for the company 'Wipro' along with the cities they live in.
ii) Find the names of the persons who do not work for 'Infosys'.
iii) Find the people whose salaries are more than that of all of the 'oracle' employees.
iv) Find the persons who works and lives in the same city.

b) Differentiate between DDL, DML, and DCL in SQL. Provide examples for each category and explain their roles in database management. **(4)**

c) Explain the various inner join operations in relational algebra with examples. **(3)**

P.T.O.